

Meriem Smati

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EDUCATION

INSA Lyon and Polytechnique Montreal	Villeurbanne, France; Montreal, Canada
<i>Ongoing Computer Science PhD degree Industrial Engineering PhD degree</i>	<i>October 2023 - Currently</i>
Higher School of Computer Science 08 May 1945 - Sidi Bel Abbes (ESI SBA)	Sidi Bel Abbes, Algeria
<i>Systems engineering Master degree Engineer Diploma in Systems engineering</i>	<i>July 2023</i>
Mohamed El Bachir El Ibrahimi University	Bordj Bou Arreridj, Algeria
<i>License degree in Information Systems and Software Engineering</i>	<i>September 2020</i>
Awards/Honors: Valedictorian, Class of 2020	
Abdelhamid Akrouf High School	Bordj Bou Arreridj, Algeria
<i>Baccalaureate in Scientific field - graduation with honors</i>	<i>July 2017</i>

WORK EXPERIENCE

INSA Lyon and Polytechnique Montreal	Villeurbanne, France; Montreal, Canada
<i>PhD Student in a dual-degree program in Computer Science and Industrial Engineering</i>	<i>October 2023 - Currently</i>
Thesis Title: "Definition and Development of System of Systems Digital Twins to improve System of Systems Resilience"	
Lumière Lyon2 University	Lyon, France
<i>Adjunct Lecturer in Computer Science</i>	<i>September 2023 - July 2024</i>
"Object Oriented Programming in Java" Course "Basics of Programming and algorithms in Python" Course "Software Development (Conception, Resource and Time Management)" Course "Management of Databases" Course "Web Automated Data Collection" Course	
Lumière Lyon2 University	Lyon, France
<i>Intern - Final Year Project</i>	<i>February-July 2023</i>
- Master Thesis Title: "Cognitive Digital Twins for IoT Data Faults Prevention" - Engineer Thesis Title: "The Use of Cognitive Digital Twins on an IoT System for Edge Anomaly Detection"	
Condor Electronics DSI	Bordj Bou Arreridj, Algeria
<i>Intern</i>	<i>September 2021</i>
Algérie Télécom	Bordj Bou Arreridj, Algeria
<i>Intern</i>	<i>March-April 2018</i>

LEADERSHIP EXPERIENCE & ACTIVITIES

International Federation of Automatic Control World Congress (IFAC)	Busan Republic of Korea, Portugal
<i>Authorship and Presentation</i>	<i>August 2025</i>
Published article: "Digital Twins of Systems of Systems: A Systematic Literature Review"	
EDT Seminar "Cas d'usage et Fabrique des Jumeaux Numériques"	Lyon, France
<i>Presentation of a poster and participation in the organization</i>	<i>January 2026</i>
Poster title: "System of Systems Digital Twins"	
Computers & Industrial Engineering Journal [ISSN 1879-0550]	Online
<i>Authorship and Publication</i>	<i>January 2026</i>
- Article title: "Enhancing data anomaly prediction and real-time physical problem detection with Digital Twins and Cognitive Super Digital Twins" - DOI: https://doi.org/10.1016/j.cie.2025.111616	

- Article title: "Introducing Digital Twins to improve the resilience of System of Systems - a Systematic Literature Review"
- DOI: <https://doi.org/10.1007/s12008-025-02460-x>

PLM25

Seville, Spain

Participation in the DT Contest

July 2025

- Report Title: "WasteTwin: A Digital Twin of a Smart Waste Collection System of Systems"
- Video Link: <https://www.intranet.disp-lab.fr/s/Y2narD4xr44GZCY>
- **Honors:** First place winner in the DT Contest

CIGI-QUALITA-MOSIM25

Troyes, France

Authorship and Presentation in CIGI Conference

July 2025

- Presentation Title: "A Smart City System of Systems Digital Twin: A Resilience-Oriented Application"

GDR GPL25

Pau, France

Presentation in the national days of GDR GPL – CNRS

June 2025

- Presentation Title: "Developing a Digital Twin for System of Systems Resilience"
- **Honors:** Best Poster Award

MODELSWARD25

Porto, Portugal

Authorship and Presentation in MODELSWARD Conference

April 2025

- Published article: "Digital Twin System of Systems: A Layered Architecture Proposal"

PLM24

Bangkok, Thailand

Authorship and Presentation in PLM Conference

July 2024

- Published article: "Enhancing Data Integrity with CSDT: A Solution to Predict Data Disturbance in IoT Systems"
- **Honors:** IFIP Marco Garetti Best Paper Award

Participation in PLM Doctoral Workshop

July 2024

- Paper title: "The Use of Digital Twins to Bolster the Resilience of an IoT System of Systems Infrastructure"
- Accomplished peer-review responsibilities
- **Honors:** IFIP Best Student Award Finalist

Participation in the DT Contest

July 2024

- Report Title: "CSDT in collaboration with DT: A solution for Disturbance Prediction in IoT Systems"
- Video Link: <https://www.intranet.disp-lab.fr/s/arbqpMqesnBtACp>

SAGIP24

Villeurbanne, France

Authorship, Presentation and participation in the organization of SAGIP Congress

May 2024

- Published article: "Enhancing Data Integrity: A Solution to Predict Data Disturbance in IoT Systems"

Generis-Publishing and Libro-Terra Book

Online

Publication of a book on Generis-Publishing and Libro-Terra

February 2024

- Published book: "Cognitive Digital Twins for Smart Manufacturing and IoT Systems"

COC'2022

Online

Authorship, Presentation in Connected Objects and Systems Conference

May 2022

- Published article: "Computer Vision and Deep Learning-Based support system for Visually Impaired People in an outdoor environment"

Developer of an "ArtificialEye" project

- a Computer Vision and Deep Learning-Based support system for Visually Impaired People

April-July 2022

Web Developer of "LECTIO" An e-book Website

January-March 2022

Project Manager of "HYGIEIA" project A health web application

April-July 2021

Member in a scientific Group Alphabit

2021

Web developer of "ARTIFEX" project An online art gallery

May-September 2020

Start-up MédiTricks a communication agency

2019-2020

Co-organizer in GDG BBA & WTM BBA Google Developers Group - Women Techmakers

2019-2021

SKILLS & INTERESTS

IT Skills

- Digital Twins
- Machine Learning Development.
- Backend Web Development
- Problem Solving
- Algorithms and Data Structures
- IoT Development.
- **Programming Languages:** Python, Java, C, Pascal, Javascript, ExpressJS (NodeJS), PHP, Laravel, GDScript
- MySQL, Oracle SQL, Docker, Godot Engine, Anylogic

Soft Skills

- Sense of communication, listening, organization and team spirit.
- Optimistic, punctual, determined, flexible and dynamic.
- Leadership and project management.
- Public Speaking.

Languages: English, French, Arabic.

Interests:

- Digital Twins, Data mining and Data analytics.
- Oceanography and marine biology.
- Traveling, painting, art, sewing, reading and learning.

CONTACTS

Portfolio: <https://meriemsmt.github.io/>

Linkedin: <https://www.linkedin.com/in/meriem-smati/>

Github: <https://github.com/meriemsmt>